

## RAUFARHOFN, Iceland

WMO: 040770

Lat: 66.456N

Lon: 15.953W

Elev: 10

StdP: 101.20

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -11.3      | -9.5 | -16.0                         | 0.9 | -10.9 | -13.9 | 1.1 | -9.2 | 21.5                  | -3.8 | 19.7 | -2.9 | 7.0                   | 230  | 0.911 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 4.3                       | 16.5            | 12.3 | 14.6 | 11.4 | 13.3 | 10.6 | 12.8                | 15.7 | 11.7 | 14.2 | 10.9 | 12.8 | 5.8                     | 230  |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 11.1                            | 8.2 | 14.3 | 10.2 | 7.8 | 12.9 | 9.6 | 7.4 | 11.9 | 36.3          | 15.8 | 33.5 | 14.4 | 31.5 | 12.9 | 17.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 17.6          | 14.7 | 13.0 | DB                         | -15.6 | 20.7               | 1.9 | 2.0                                                | -16.9 | 22.2       | -18.0 | 23.3       | -19.1 | 24.5       | -20.4 | 26.0 | (4) |
| (5)               |      |      | WB                         | -16.4 | 15.0               | 1.9 | 1.1                                                | -17.7 | 15.8       | -18.9 | 16.5       | -19.9 | 17.2       | -21.3 | 18.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 3.0    | -1.0  | -1.5  | -1.1  | 1.0   | 3.5   | 6.8   | 9.1   | 9.3   | 6.8   | 3.1   | 0.5   | -0.4 |
|                                                                                  | DBStd       | 5.09   | 3.44  | 3.78  | 3.91  | 3.52  | 2.96  | 2.85  | 2.55  | 2.27  | 3.11  | 3.35  | 3.57  | 3.75 |
|                                                                                  | HDD10.0     | 2595   | 342   | 321   | 343   | 269   | 203   | 103   | 47    | 42    | 103   | 215   | 286   | 322  |
|                                                                                  | HDD18.3     | 5582   | 600   | 554   | 602   | 519   | 461   | 346   | 285   | 281   | 346   | 473   | 536   | 580  |
|                                                                                  | CDD10.0     | 57     | 0     | 0     | 0     | 1     | 1     | 7     | 21    | 19    | 8     | 1     | 0     | 0    |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 6.4    | 8.0   | 7.8   | 7.3   | 6.1   | 5.6   | 5.4   | 5.0   | 5.0   | 5.6   | 6.8   | 6.7   | 7.2  |
| Precipitation                                                                    | PrecAvg     | 649    | 61    | 51    | 57    | 40    | 34    | 38    | 47    | 51    | 67    | 79    | 71    | 58   |
|                                                                                  | PrecMax     | 866    | 100   | 109   | 106   | 61    | 70    | 101   | 115   | 99    | 129   | 132   | 119   | 124  |
|                                                                                  | PrecMin     | 466    | 24    | 20    | 18    | 13    | 6     | 5     | 15    | 8     | 17    | 13    | 30    | 23   |
|                                                                                  | PrecStd     | 100    | 22    | 26    | 22    | 11    | 18    | 22    | 27    | 27    | 32    | 32    | 21    | 25   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.4   | 7.9   | 9.2   | 13.4  | 14.2  | 18.5  | 19.3  | 18.9  | 16.6  | 13.1  | 10.6  | 10.0 |
|                                                                                  |             | MCWB   | 5.5   | 6.2   | 6.8   | 9.2   | 9.8   | 12.6  | 13.7  | 12.8  | 12.5  | 9.8   | 8.7   | 7.2  |
|                                                                                  | 2%          | DB     | 5.7   | 6.1   | 6.9   | 9.8   | 11.5  | 14.7  | 16.2  | 15.7  | 14.2  | 10.7  | 7.1   | 7.5  |
|                                                                                  |             | MCWB   | 4.0   | 4.6   | 4.7   | 7.0   | 8.4   | 10.6  | 12.3  | 12.5  | 10.9  | 8.4   | 5.5   | 5.5  |
|                                                                                  | 5%          | DB     | 4.6   | 4.8   | 5.2   | 7.4   | 9.6   | 12.6  | 14.2  | 14.0  | 12.5  | 8.9   | 5.9   | 5.6  |
|                                                                                  |             | MCWB   | 3.3   | 3.4   | 3.6   | 5.3   | 7.1   | 9.6   | 11.4  | 11.5  | 9.9   | 7.5   | 4.9   | 4.2  |
|                                                                                  | 10%         | DB     | 3.5   | 3.7   | 4.0   | 5.7   | 7.9   | 11.1  | 12.8  | 12.7  | 11.0  | 7.6   | 5.1   | 4.5  |
|                                                                                  |             | MCWB   | 2.5   | 2.6   | 2.6   | 4.2   | 6.2   | 8.7   | 10.6  | 10.7  | 9.3   | 6.6   | 4.3   | 3.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 5.7   | 6.3   | 6.8   | 9.2   | 10.4  | 13.0  | 14.2  | 13.9  | 13.3  | 10.1  | 8.8   | 7.2  |
|                                                                                  |             | MCDB   | 6.9   | 7.7   | 9.2   | 12.9  | 13.8  | 17.1  | 18.0  | 17.1  | 16.0  | 12.6  | 10.7  | 9.6  |
|                                                                                  | 2%          | WB     | 4.2   | 4.7   | 4.9   | 7.0   | 8.6   | 11.1  | 12.8  | 12.7  | 11.2  | 8.6   | 6.0   | 5.5  |
|                                                                                  |             | MCDB   | 5.4   | 6.0   | 6.5   | 9.7   | 11.1  | 14.0  | 15.7  | 15.2  | 13.4  | 10.3  | 6.8   | 7.1  |
|                                                                                  | 5%          | WB     | 3.4   | 3.6   | 3.8   | 5.4   | 7.2   | 9.7   | 11.6  | 11.7  | 10.3  | 7.7   | 5.1   | 4.5  |
|                                                                                  |             | MCDB   | 4.4   | 4.6   | 5.2   | 7.2   | 9.2   | 12.2  | 13.9  | 13.8  | 12.2  | 8.8   | 5.7   | 5.6  |
|                                                                                  | 10%         | WB     | 2.6   | 2.7   | 2.7   | 4.3   | 6.2   | 8.8   | 10.7  | 10.8  | 9.3   | 6.6   | 4.4   | 3.5  |
|                                                                                  |             | MCDB   | 3.5   | 3.6   | 3.8   | 5.7   | 7.8   | 10.9  | 12.6  | 12.4  | 10.8  | 7.5   | 5.1   | 4.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.4   | 4.7   | 4.5   | 4.5   | 4.2   | 4.4   | 4.1   | 4.3   | 4.5   | 3.9   | 4.2   | 4.6  |
|                                                                                  |             | MCDBR  | 5.1   | 5.7   | 5.8   | 7.4   | 7.6   | 8.1   | 7.5   | 7.2   | 6.9   | 6.0   | 4.9   | 5.8  |
|                                                                                  |             | MCWBR  | 4.6   | 5.4   | 4.9   | 5.7   | 5.5   | 5.0   | 4.5   | 4.9   | 4.7   | 4.7   | 4.7   | 5.3  |
|                                                                                  | 5% WB       | MCDBR  | 4.8   | 5.5   | 5.6   | 6.7   | 7.2   | 7.5   | 6.8   | 6.5   | 6.5   | 5.5   | 4.3   | 5.5  |
|                                                                                  |             | MCWBR  | 4.5   | 5.3   | 4.8   | 5.3   | 5.4   | 4.9   | 4.3   | 4.6   | 4.7   | 4.3   | 4.2   | 5.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.377  | 0.296 | 0.322 | 0.351 | 0.355 | 0.344 | 0.348 | 0.342 | 0.333 | 0.301 | 0.370 | 0.382 |      |
|                                                                                  | taud        | 2.536  | 2.173 | 2.303 | 2.312 | 2.385 | 2.485 | 2.505 | 2.530 | 2.463 | 2.233 | 2.523 | 2.551 |      |
|                                                                                  | Ebn at Noon | 184    | 540   | 734   | 807   | 844   | 866   | 850   | 819   | 730   | 522   | 169   | 33    |      |
|                                                                                  | Edh at Noon | 19     | 52    | 76    | 97    | 100   | 93    | 88    | 78    | 66    | 48    | 18    | 8     |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.06   | 0.36  | 1.27  | 2.74  | 4.00  | 4.86  | 4.13  | 3.12  | 1.76  | 0.59  | 0.11  | 0.01  |      |
|                                                                                  | RadStd      | 0.01   | 0.05  | 0.17  | 0.27  | 0.38  | 0.44  | 0.37  | 0.26  | 0.12  | 0.06  | 0.01  | 0.00  |      |

## Historical Trends

|                             |  | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|-------|---------|-------|------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)     | (m)     | (n)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (7 neighbors) |  | +0.37 | +1.04   | N/A    | N/A   | N/A     | N/A   | -118 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page